Geometric Transformation

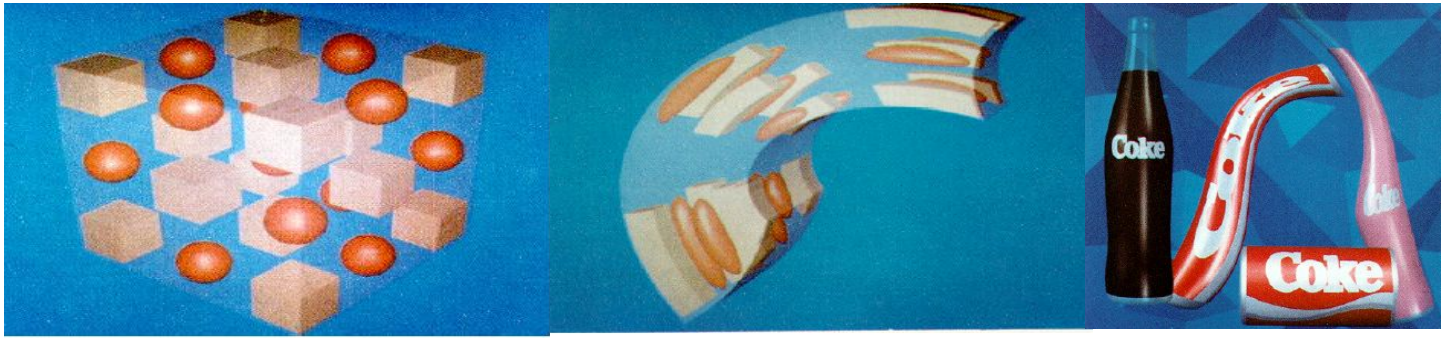
What is a Transformation

- Transformation:
  - An operation that changes one configuration into another
- For images, shapes, *etc.*
  - A geometric transformation maps positions that define the object to other positions
  - Linear transformation means the transformation is defined by a linear function... which is what matrices are good for.

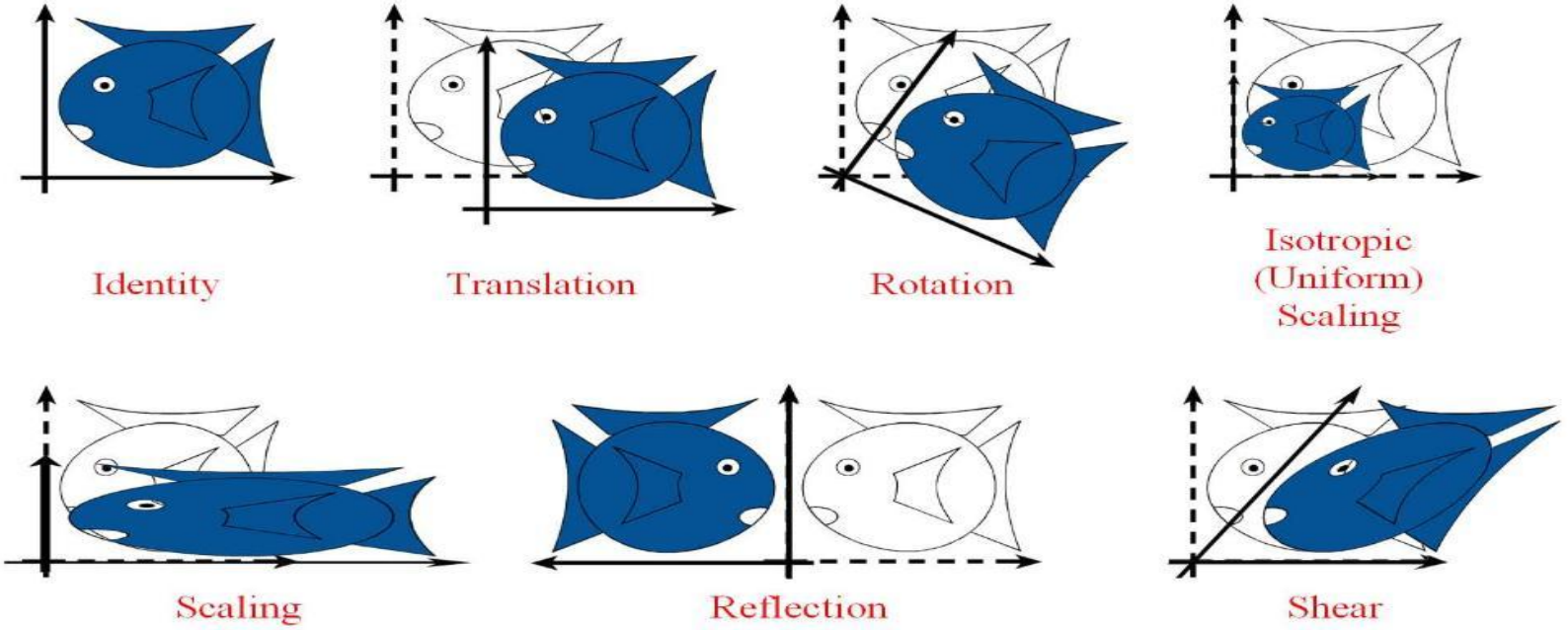
What is a Transformation?

- A function that maps points x to points x':

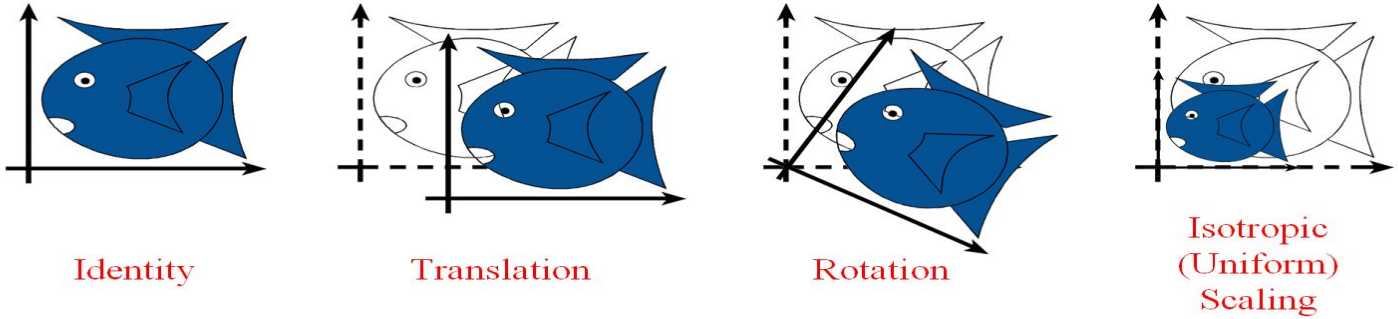
  Applications: animation, deformation, viewing, projection, real-time shadows, …



Examples of Transformation



Simple Transformations



- Can be combined
- Are these operations invertible?

Yes, except scale = 0

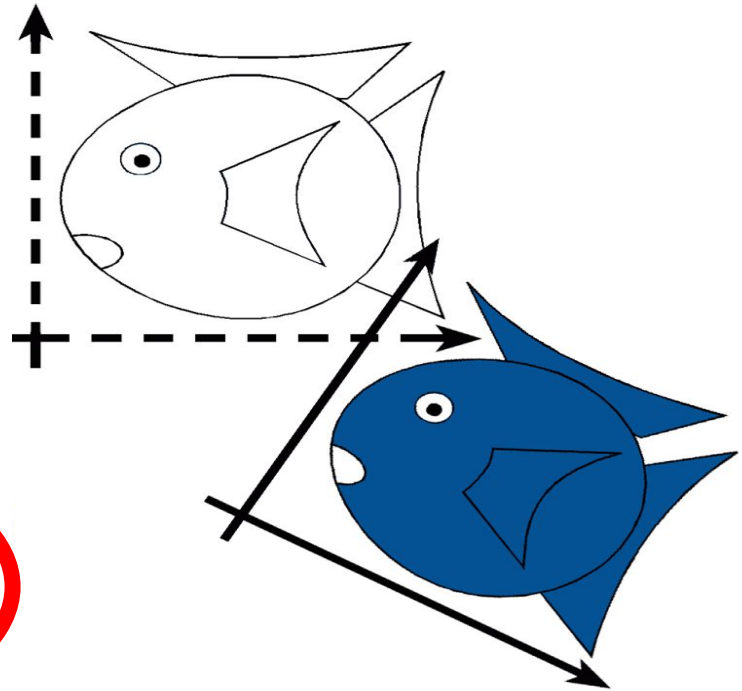
Rigid-Body / Euclidean Transforms

- Preserves distances
- Preserves angles

Rigid / Euclidean

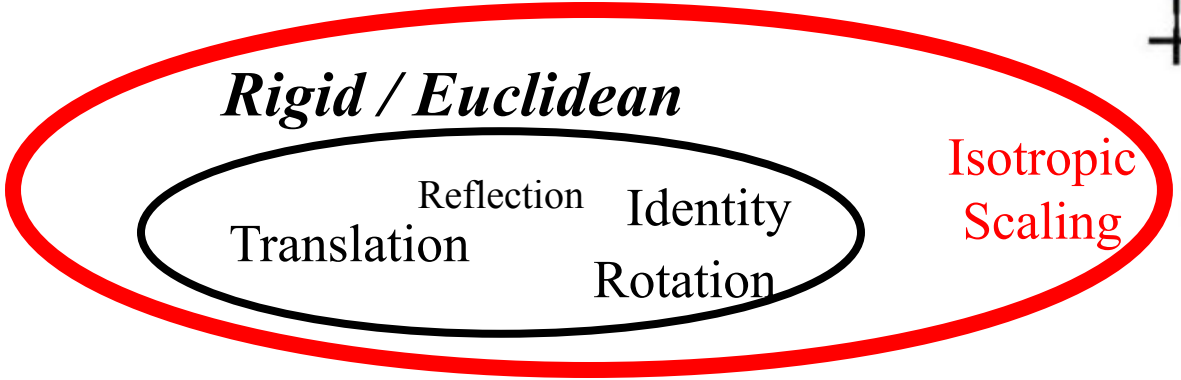
Translation

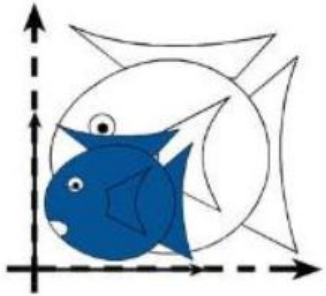
Identity
Rotation
Reflection



Similitudes / Similarity Transforms

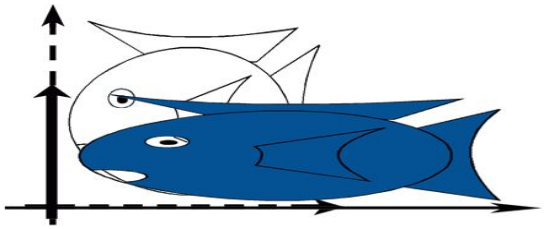
- Preserves angles



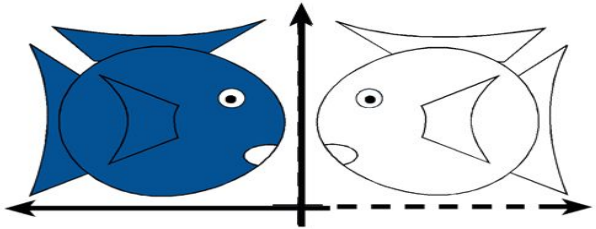


Isotropic
(Uniform)
Scaling

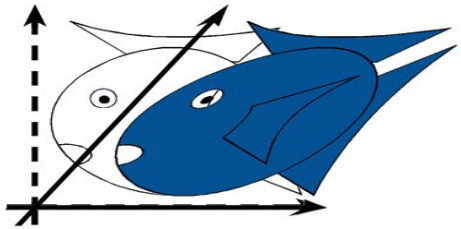
Linear Transformations



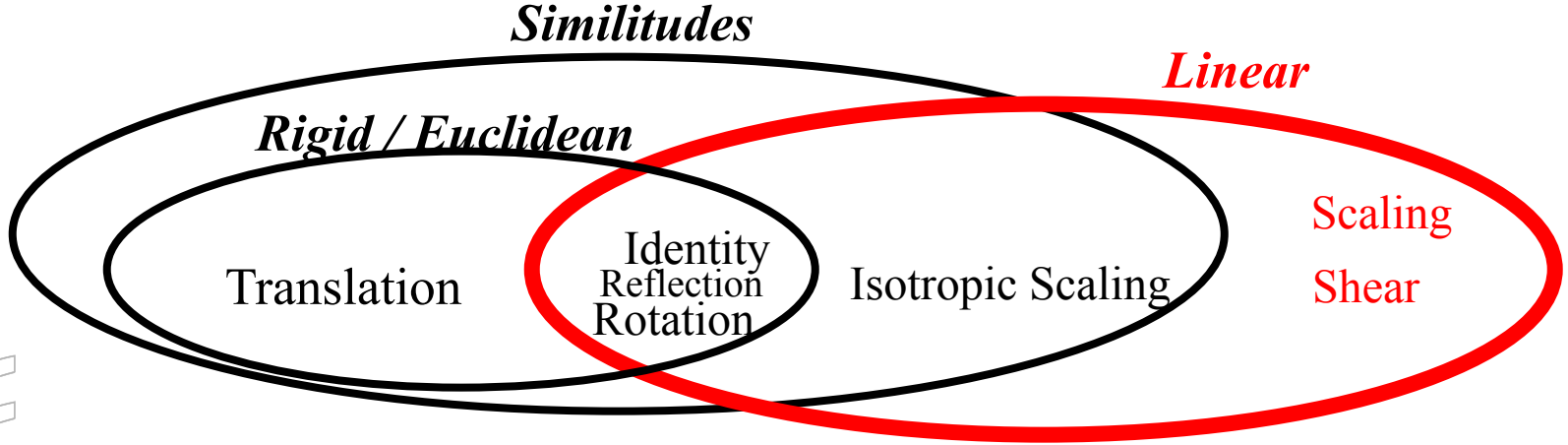
Scaling



Reflection



Shear

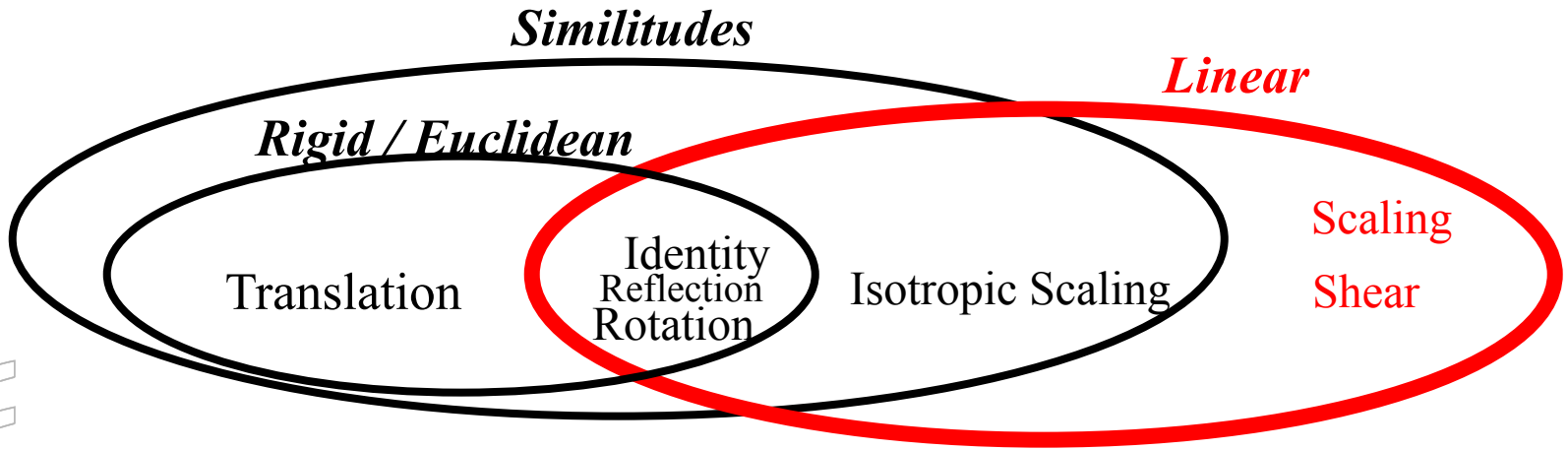


Linear Transformations

Additivity: *T(u + v) = T(u) + T(v) , here u and v are* ***vectors***

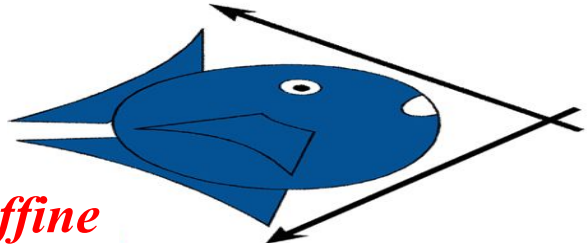
Homogeneity**:** *T(au) = a T(u)*

Q: Translation: T(u) = u + t, Linear??

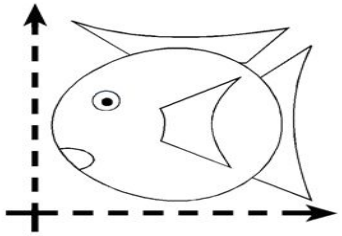


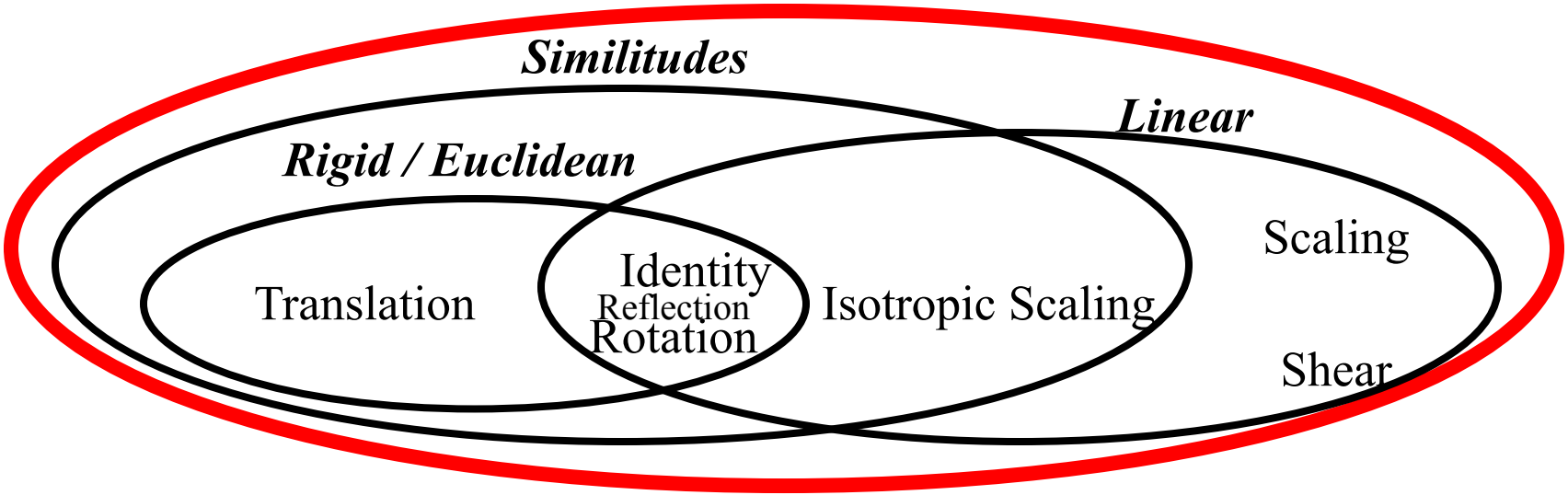
Affine Transformations

- preserves parallel lines (parallelism)

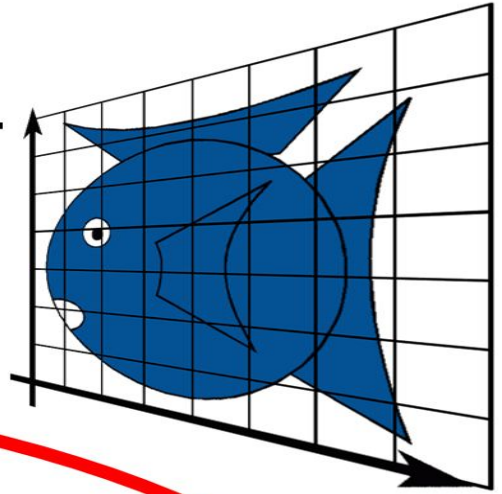


Affine

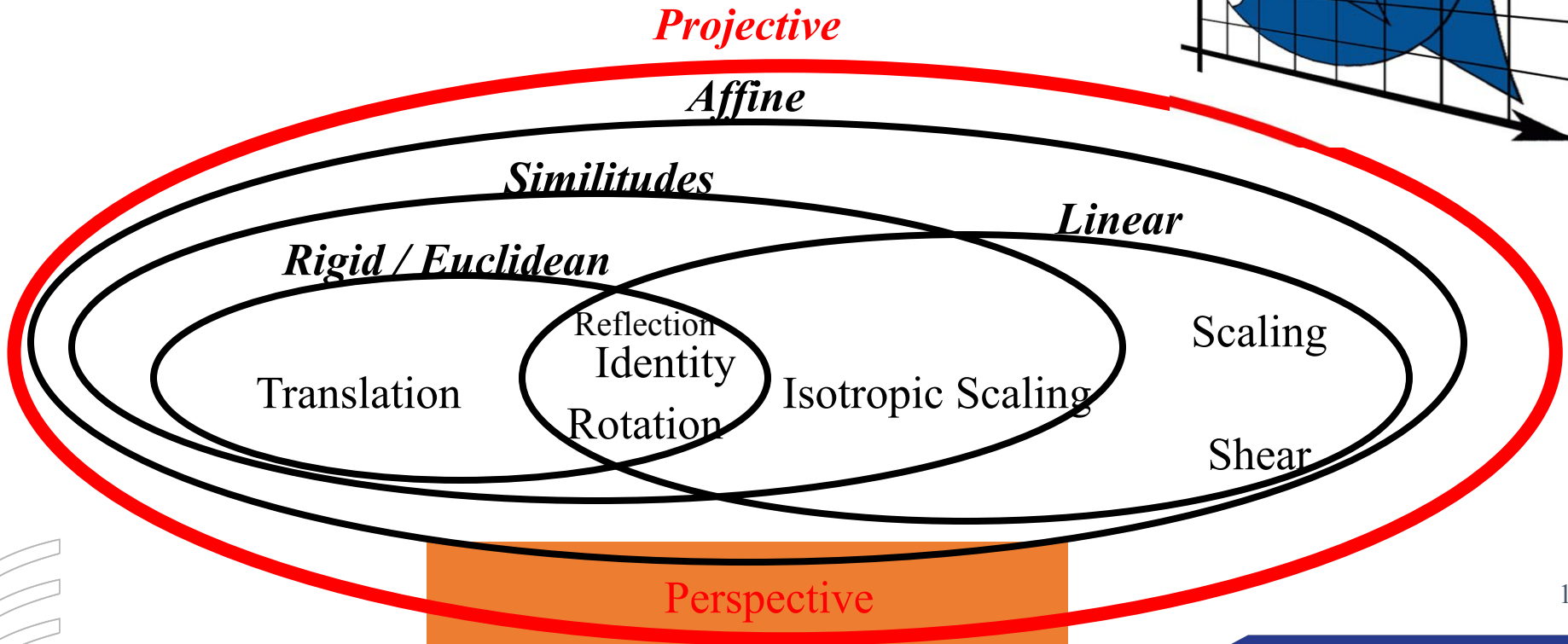




Projective Transformations



- preserves lines (Collinearity)



How are Transforms Represented?

$$x' = ax + by + c$$

$$y' = dx + ey + f$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ d & e \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} c \\ f \end{bmatrix}$$

$$p' = M p + t$$

Translation in homogeneous coordinates

$$x' = ax + by + c$$

$$y' = dx + ey + f$$

Cartesian formulation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{pmatrix} a & b \\ d & e \end{pmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} c \\ f \end{bmatrix}$$

$$p' = M\,p + t$$

Homogeneous formulation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{pmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{pmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$p' = M\,p$$

Homogeneous Coordinates

- Translation, scaling and rotation are expressed (non-homogeneously) as:
  - translation: $P' = P + T$
  - Scale: $P' = S \cdot P$
  - Rotate: $P' = R \cdot P$
- Composition is difficult to express, since translation not expressed as a matrix multiplication
- Homogeneous coordinates allow all three to be expressed homogeneously, using multiplication by 3×3 matrices
- W is 1 for affine transformations in graphics

Homogeneous Coordinates

- Add an extra dimension
  - in 2D, we use 3 x 3 matrices
  - In 3D, we use 4 x 4 matrices
- Each point has an extra value, w

$$\begin{pmatrix} x' \\ y' \\ z' \\ w' \end{pmatrix} = \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix}$$

$$p' = M\,p$$

Homogeneous Coordinates

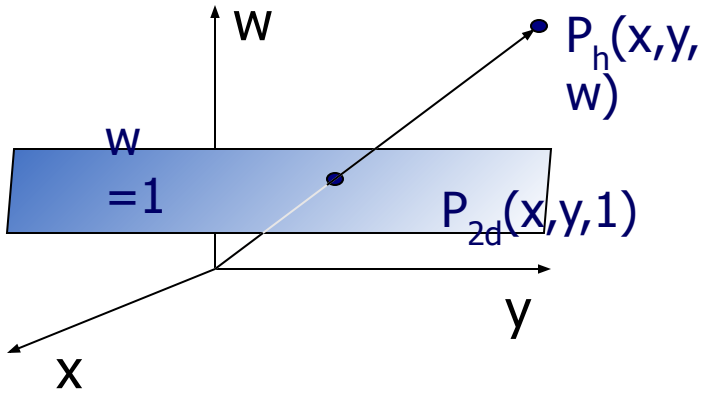
- Most of the time w = 1, and we can ignore it

$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

- If we multiply a homogeneous coordinate by an *affine matrix*, w is unchanged

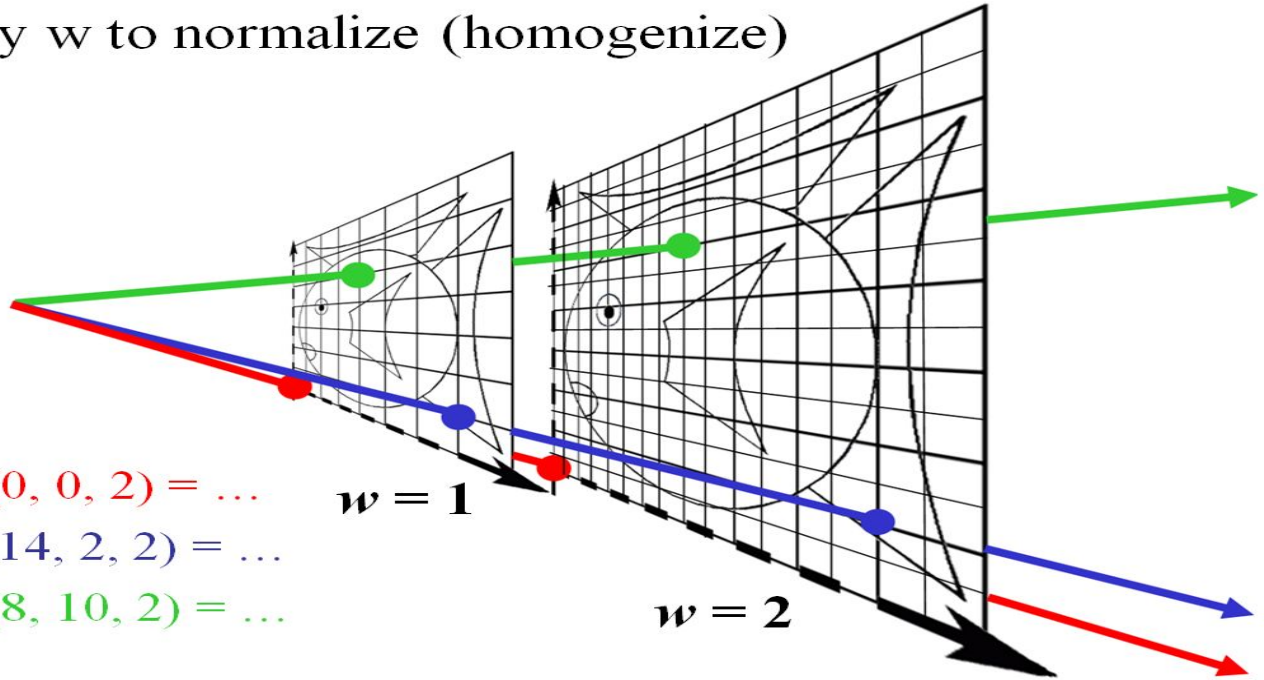
Homogeneous Coordinates

- P_{2d} is a projection of P_h onto the $w = 1$ plane
- So an infinite number of points correspond to : they constitute the whole line (tx, ty, tw)



Homogeneous Visualization

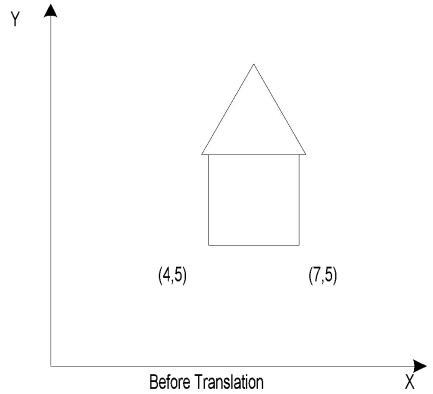
- Divide by w to normalize (homogenize)
- W = 0?

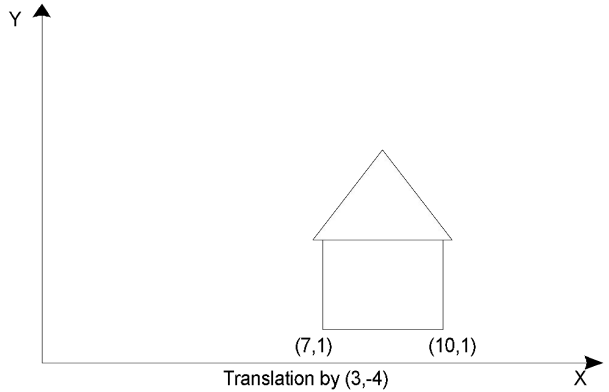


(0, 0, 1) = (0, 0, 2) = …
(7, 1, 1) = (14, 2, 2) = …
(4, 5, 1) = (8, 10, 2) = …

Mechanics of Transformations

Translation – 2D





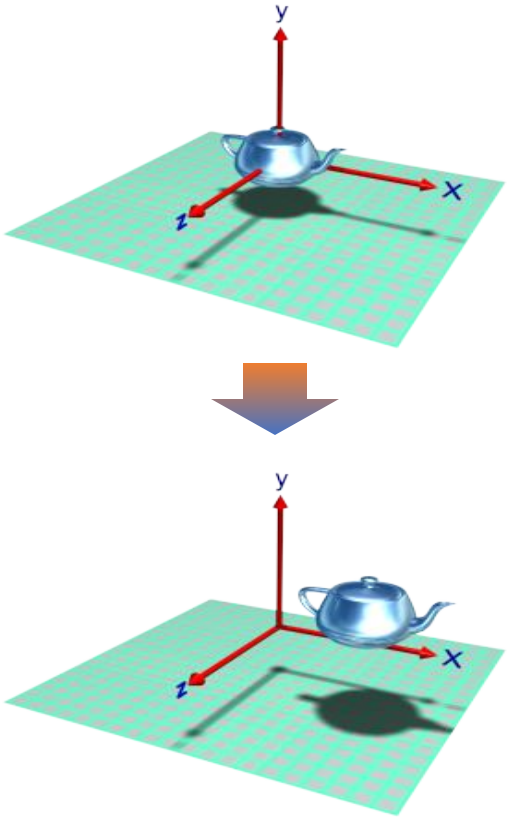
$x' = x + d_x$
$y' = y + d_y$

$$P = \begin{bmatrix} x \\ y \end{bmatrix} \quad P' = \begin{bmatrix} x' \\ y' \end{bmatrix} \quad T = \begin{bmatrix} d_x \\ d_y \end{bmatrix} \qquad P' = P + T$$

Homogeniou s Form

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & d_x \\ 0 & 1 & d_y \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Translation – 3D



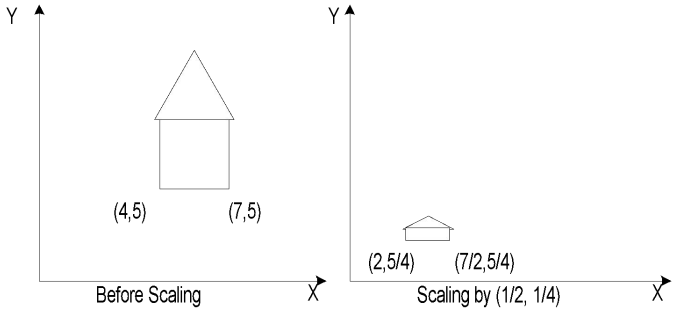
$$x' = x + d_x$$
$$y' = y + d_y$$
$$z' = z + d_z$$

$$\begin{bmatrix} 1 & 0 & 0 & d_x \\ 0 & 1 & 0 & d_y \\ 0 & 0 & 1 & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x + d_x \\ y + d_y \\ z + d_z \\ 1 \end{bmatrix}$$

$$\Downarrow \qquad \Downarrow \qquad \Downarrow$$

$$T(d_x, d_y, d_z) \quad * \quad P \quad = \quad P'$$

Scaling – 2D



Types of Scaling:

- Differential (s_x != s_y)
- Uniform (s_x = s_y)

$$x' = s_x * x$$

$$y' = s_y * y$$

$$S \quad * \quad P \quad = \quad P'$$

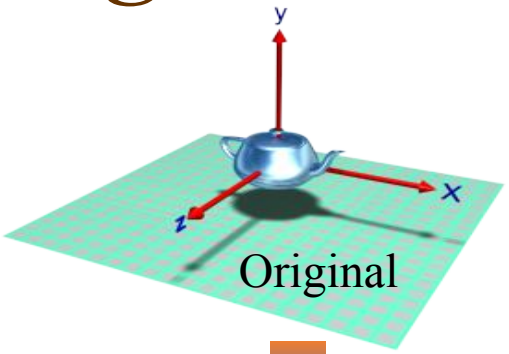
$$\Downarrow \qquad \Downarrow \qquad \Downarrow$$

$$\begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} * \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x * s_x \\ y * s_y \end{bmatrix}$$

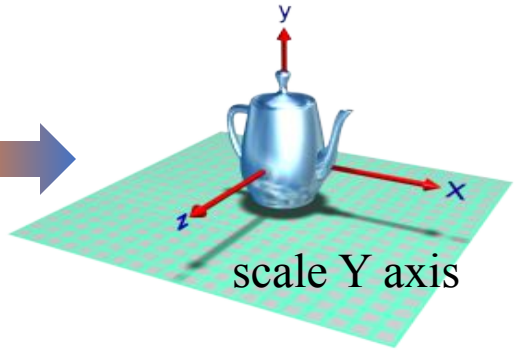
Homogenious Form

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

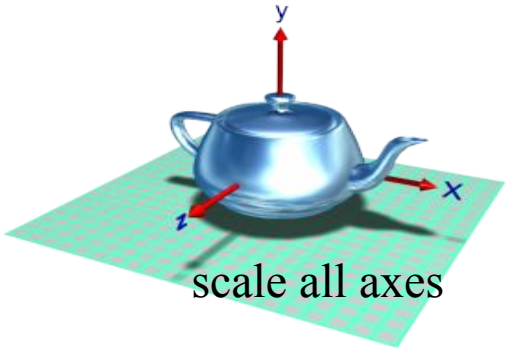
Scaling – 3D



Original



scale Y axis



scale all axes

$$x' = s_x * x$$
$$y' = s_y * y$$
$$z' = s_z * z$$

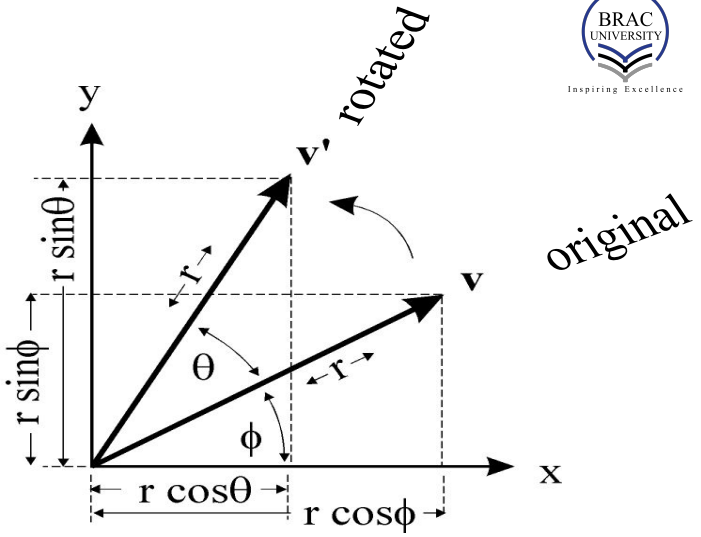
$$S(s_x, s_y, s_z) \quad * \quad P \quad = \quad P'$$

$$\begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x * s_x \\ y * s_y \\ z * s_z \\ 1 \end{bmatrix}$$

Rotation – 2D

$$\mathbf{v} = \begin{bmatrix} r\cos\phi \\ r\sin\phi \end{bmatrix}$$

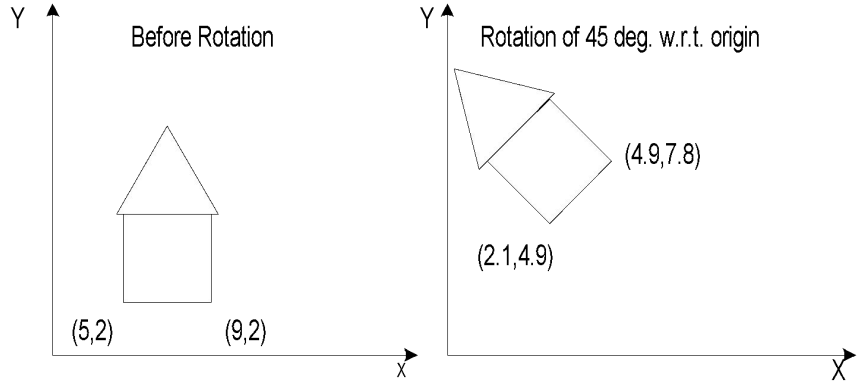
$$\mathbf{v}' = \begin{bmatrix} r\cos(\phi+\theta) \\ r\sin(\phi+\theta) \end{bmatrix}$$



$$\text{expand}\,(\phi+\theta) \Rightarrow \begin{cases} x' = r\cos\phi\cos\theta - r\sin\phi\sin\theta \\ y' = r\cos\phi\sin\theta + r\sin\phi\cos\theta \end{cases}$$

$$\text{but} \quad \begin{matrix} x = r\cos\phi \\ y = r\sin\phi \end{matrix} \Rightarrow \begin{matrix} x' = x\cos\theta - y\sin\theta \\ y' = x\sin\theta + y\cos\theta \end{matrix}$$

Rotation – 2D



$$x*\cos\theta - y*\sin\theta = x'$$

$$x*\sin\theta + y*\cos\theta = y'$$

$$R \quad * \quad P = \quad P'$$

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} * \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x*\cos\theta - y*\sin\theta \\ x*\sin\theta + y*\cos\theta \end{bmatrix}$$

Homogenious Form

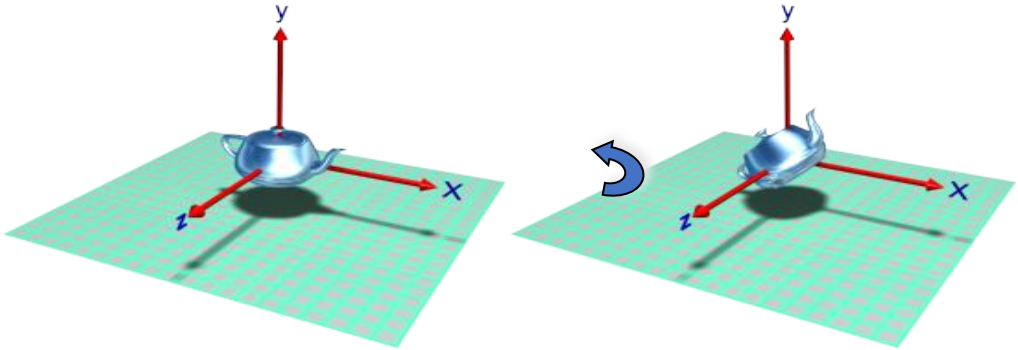
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Rotation – 3D

For 3D-Rotation 2 parameters are needed

- Angle of rotation
- Axis of rotation

Rotation about z-axis:

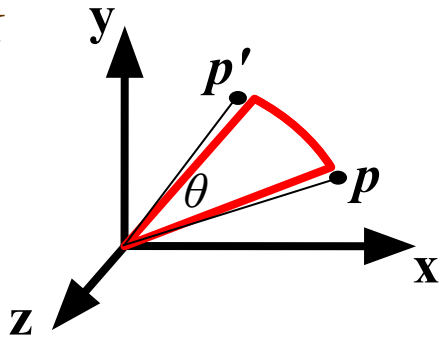


$$R_{\theta,k} \quad * \quad P \quad = \quad P'$$

$$\Downarrow \qquad\qquad \Downarrow \qquad\qquad \Downarrow$$

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x * \cos\theta - y * \sin\theta \\ x * \sin\theta + y * \cos\theta \\ z \\ 1 \end{bmatrix}$$

Rotation about Z axis: $R_{\theta,k}$



- About z axis

$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

Rotation about Y-axis & X-axis

About y-axis

$$R_{\theta,j} * P = P'$$

$$\Downarrow \quad \Downarrow \quad \Downarrow$$

$$\begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x * \cos\theta + z * \sin\theta \\ y \\ -x * \sin\theta + z * \cos\theta \\ 1 \end{bmatrix}$$

About x-axis

$$R_{\theta,i} * P = P'$$

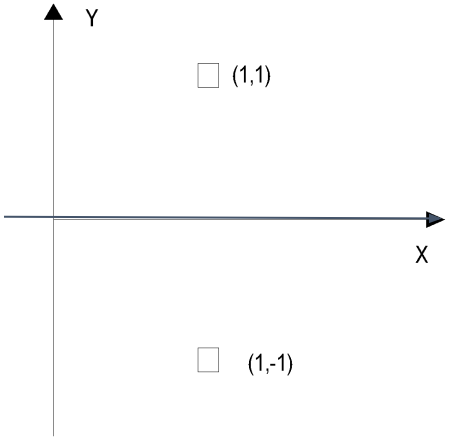
$$\Downarrow \quad \Downarrow \quad \Downarrow$$

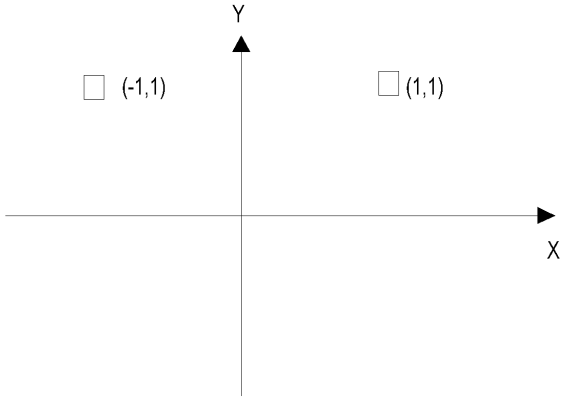
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y * \cos\theta - z * \sin\theta \\ y * \sin\theta + z * \cos\theta \\ 1 \end{bmatrix}$$

Properties of rotation matrix

- The columns of rotation matrix are unit vectors perpendicular to each other
- The column vectors indicate where the unit vectors along the principal axes are transformed
- The rows of rotation matrix are unit vectors perpendicular to each other
- The row vectors indicate the vectors that are transformed into the unit vectors along the principal axes
- The inverse of rotation matrix is its transpose

Mirror Reflection





Reflection about X - axis

$$x' = x \quad y' = -y$$

$$M_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reflection about Y - axis

$$x' = -x \quad y' = y$$

$$M_y = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Translation(d_x, d_y)

$$\begin{bmatrix} 1 & 0 & d_x \\ 0 & 1 & d_y \\ 0 & 0 & 1 \end{bmatrix}$$

Scale(s_x, s_y)

$$\begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation(θ)

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reflection$(x\ axis)$

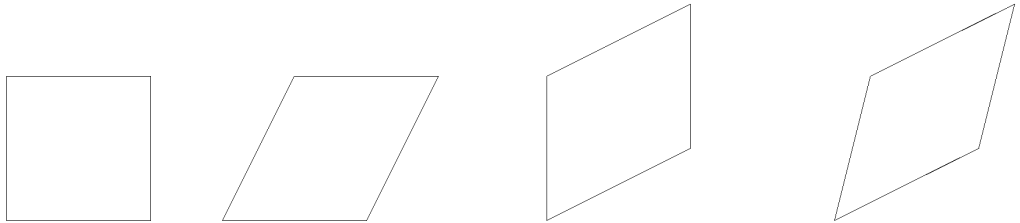
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reflection$(y\ axis)$

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Shearing Transformation

$$SH_x = \begin{bmatrix} 1 & a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad SH_y = \begin{bmatrix} 1 & 0 & 0 \\ b & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad SH_{xy} = \begin{bmatrix} 1 & a & 0 \\ b & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



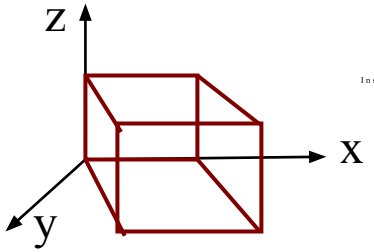
unit cube

Sheared in X direction

Sheared in Y direction

Sheared in both X and Y direction

Shear along Z-axis

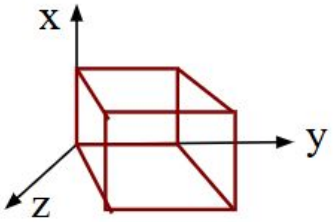


$$SH_{xy}(sh_x, sh_y) \quad * \quad P \quad = \quad P'$$

$$\Downarrow \qquad \Downarrow \qquad \Downarrow$$

$$\begin{bmatrix} 1 & 0 & sh_x & 0 \\ 0 & 1 & sh_y & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x + z * sh_x \\ y + z * sh_y \\ z \\ 1 \end{bmatrix}$$

Shear along X-axis

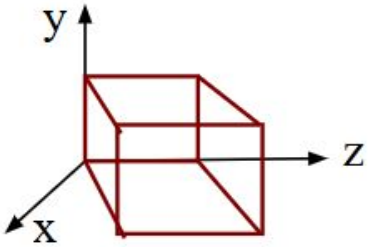


$$\begin{bmatrix} X_{new} \\ Y_{new} \\ Z_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ Sh_y & 1 & 0 & 0 \\ Sh_z & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ Z_{old} \\ 1 \end{bmatrix}$$

3D Shearing Matrix

(In X axis)

Shear along Y-axis



$$\begin{bmatrix} X_{new} \\ Y_{new} \\ Z_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & Sh_x & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & Sh_z & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ Z_{old} \\ 1 \end{bmatrix}$$

3D Shearing Matrix

(In Y axis)

Inverse Transforms

- In general: $\mathbf{A}^{-1}$ undoes effect of $\mathbf{A}$
- Special cases:
  - Translation: negate t_x and t_y
  - Rotation: transpose
  - Scale: invert diagonal (axis-aligned scales)
- Others:
  - Invert matrix
  - Invert SVD matrices

Inverse Transformations

Translation : $T^{-1}_{(dx,\ dy)} = T_{(-dx,-\ dy)}$

Rotation : $R^{-1}_{(\theta)} = R_{(-\theta)} = R^{T}_{(\theta)}$

Scaling : $S^{-1}_{(sx,\ sy)} = S_{(1/sx,\ 1/sy)}$

Mirror Ref : $M_x^{-1} = M_x$

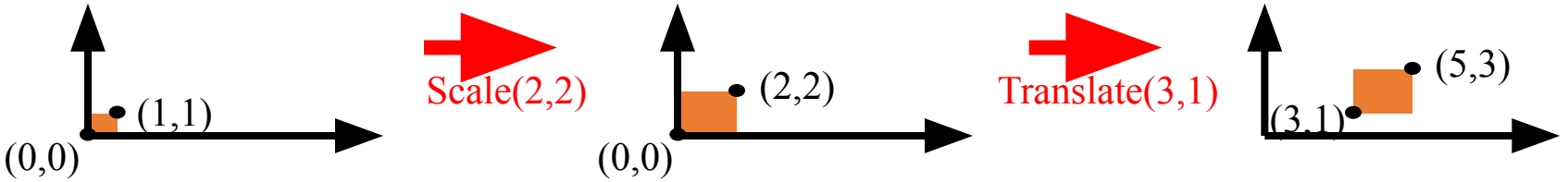
$M_y^{-1} = M_y$

Shear : $Sh_x^{-1} = Sh_{-x}$

Composing Transformations

How are transforms combined?

Scale then Translate



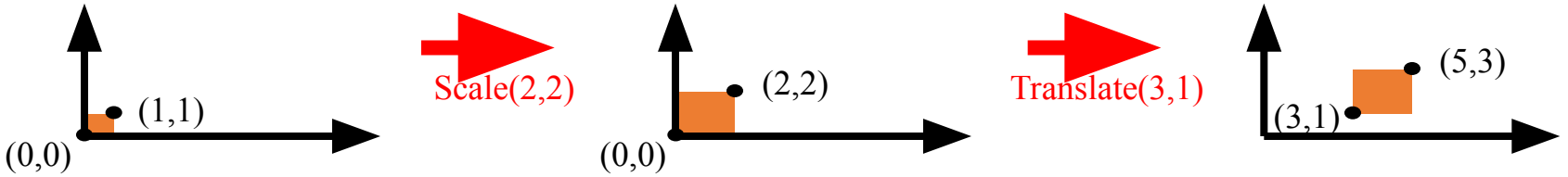
Use matrix multiplication: $p' = T(Sp) = TSp$

$$TS = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

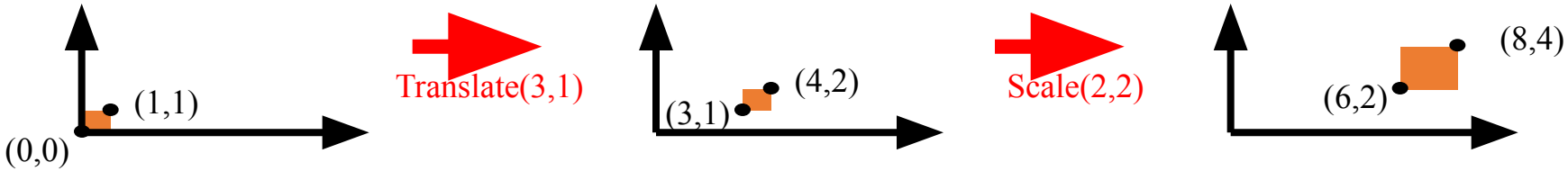
Caution: matrix multiplication is NOT commutative!

Non-commutative Composition

Scale then Translate: $p' = T(S\,p) = TS\,p$



Translate then Scale: $p' = S(T\,p) = ST\,p$



Non-commutative Composition

Scale then Translate: $p' = T(S\,p) = TS\,p$

$$TS = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Translate then Scale: $p' = S(T\,p) = ST\,p$

$$ST = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 6 \\ 0 & 2 & 2 \\ 0 & 0 & 1 \end{pmatrix}$$

Combining Translations, Rotations

- Order matters!! TR is not the same as RT (demo)
- General form for rigid body transforms
- We show rotation first, then translation (commonly used to position objects) on next slide. Slide after that works it out the other way

Rotate then Translate

$$P' = (TR)P = MP = RP + T$$

$$M = \begin{pmatrix} 1 & 0 & 0 & T_x \\ 0 & 1 & 0 & T_y \\ 0 & 0 & 1 & T_z \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} R_{11} & R_{12} & R_{13} & 0 \\ R_{21} & R_{22} & R_{23} & 0 \\ R_{31} & R_{32} & R_{33} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \left(\begin{array}{ccc|c} R_{11} & R_{12} & R_{13} & T_x \\ R_{21} & R_{22} & R_{23} & T_y \\ R_{31} & R_{32} & R_{33} & T_z \\ \hline 0 & 0 & 0 & 1 \end{array}\right) = \begin{pmatrix} R & T \\ 0 & 1 \end{pmatrix}$$

Translate then Rotate

$$P' = (RT)P = MP = R(P+T) = RP + RT$$

$$M = \begin{pmatrix} R_{11} & R_{12} & R_{13} & 0 \\ R_{21} & R_{22} & R_{23} & 0 \\ R_{31} & R_{32} & R_{33} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & T_x \\ 0 & 1 & 0 & T_y \\ 0 & 0 & 1 & T_z \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} R_{3\times3} & R_{3\times3}T_{3\times1} \\ 0_{1\times3} & 1 \end{pmatrix}$$

Associativity of Matrix Multiplication

Create new affine transformations by multiplying sequences of the above basic transformations.

$$\mathbf{q} = \mathbf{CBAp}$$

$$\mathbf{q} = ((\mathbf{CB})\mathbf{A})\mathbf{p} = (\mathbf{C}(\mathbf{BA}))\mathbf{p} = \mathbf{C}(\mathbf{B}(\mathbf{Ap})) \text{ etc.}$$

matrix multiplication is associative.

To transform just a point, better to do $\mathbf{q} = \mathbf{C}(\mathbf{B}(\mathbf{Ap}))$

But to transform many points, best to do

$$\mathbf{M} = \mathbf{CBA}$$

then do $\mathbf{q} = \mathbf{Mp}$ for any point **p** to be rendered.

For geometric pipeline transformation, define **M** and set it up with the model-view matrix and apply it to any vertex subsequently defined to its setting.

Example Composite Transforms

Rotation of θ about P(h,k): $R_{\theta,P}$

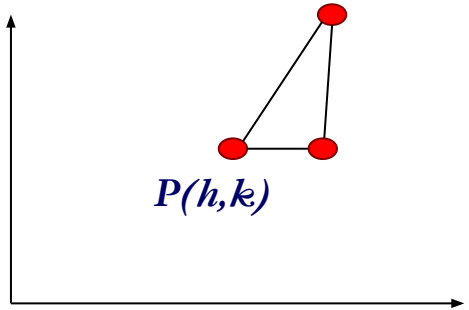
Step 1: Translate P(-h,-k) to origin

Step 2: Rotate θ w.r.t to origin

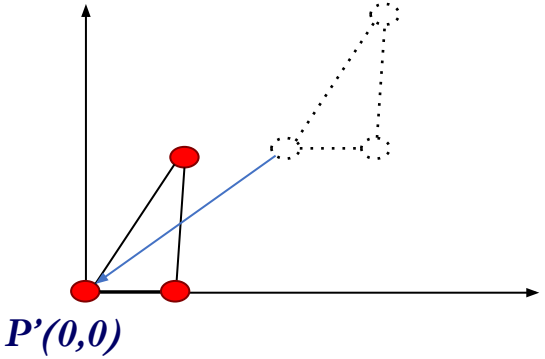
Step 3: Translate (0,0) to P(h,k)

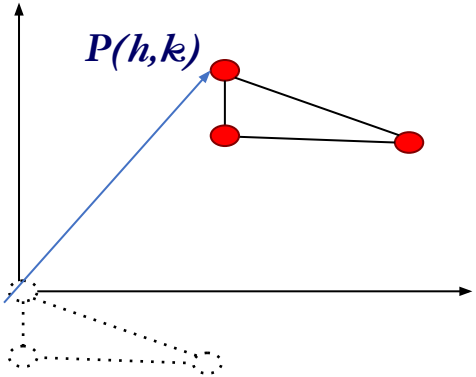
$R_{\theta,P}$= T(h ,k) * R_{θ}* T(-h ,-k)

Rotation of 90º clockwise about P(h,k)



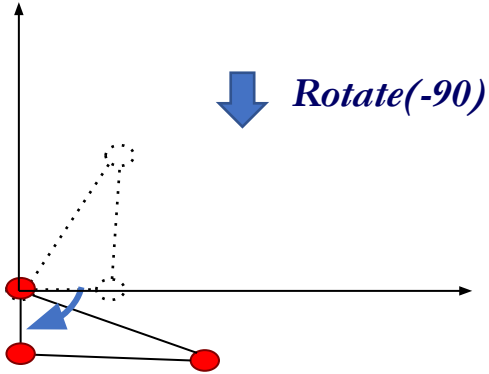
Translate
(-h,-k)





Translate
(h,k)

Rotate(-90)



$$P' = T_{(h,k)} \times R_{(\theta)} \times T_{(-h,-k)} \times P$$

$$Composite\ matrix, M = T_{(h,k)} \times R_{(\theta)} \times T_{(-h,-k)}$$

$$= \begin{bmatrix} 1 & 0 & h \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -h \\ 0 & 1 & -k \\ 0 & 0 & 1 \end{bmatrix}$$

Example 1

Suppose, a composite transformation is defined as rotating 90 degree counterclockwise with respect to point (5, 5). Calculate the composite transformation matrix in homogeneous form. Then, find the new coordinates of the point (10, 10) after transformation.

$$Composite\ matrix, M = T_{(5,5)} \times R_{(90)} \times T_{(-5,-5)}$$

$$Composite\ matrix, M = T_{(5,5)} \times R_{(90)} \times T_{(-5,-5)}$$

$$= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 90 & -\sin 90 & 0 \\ \sin 90 & \cos 90 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 & 10 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P' = M \times P$$

$$= \begin{bmatrix} 0 & -1 & 10 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 10 \\ 1 \end{bmatrix}$$

So, the new coordinate of P is (0, 10)

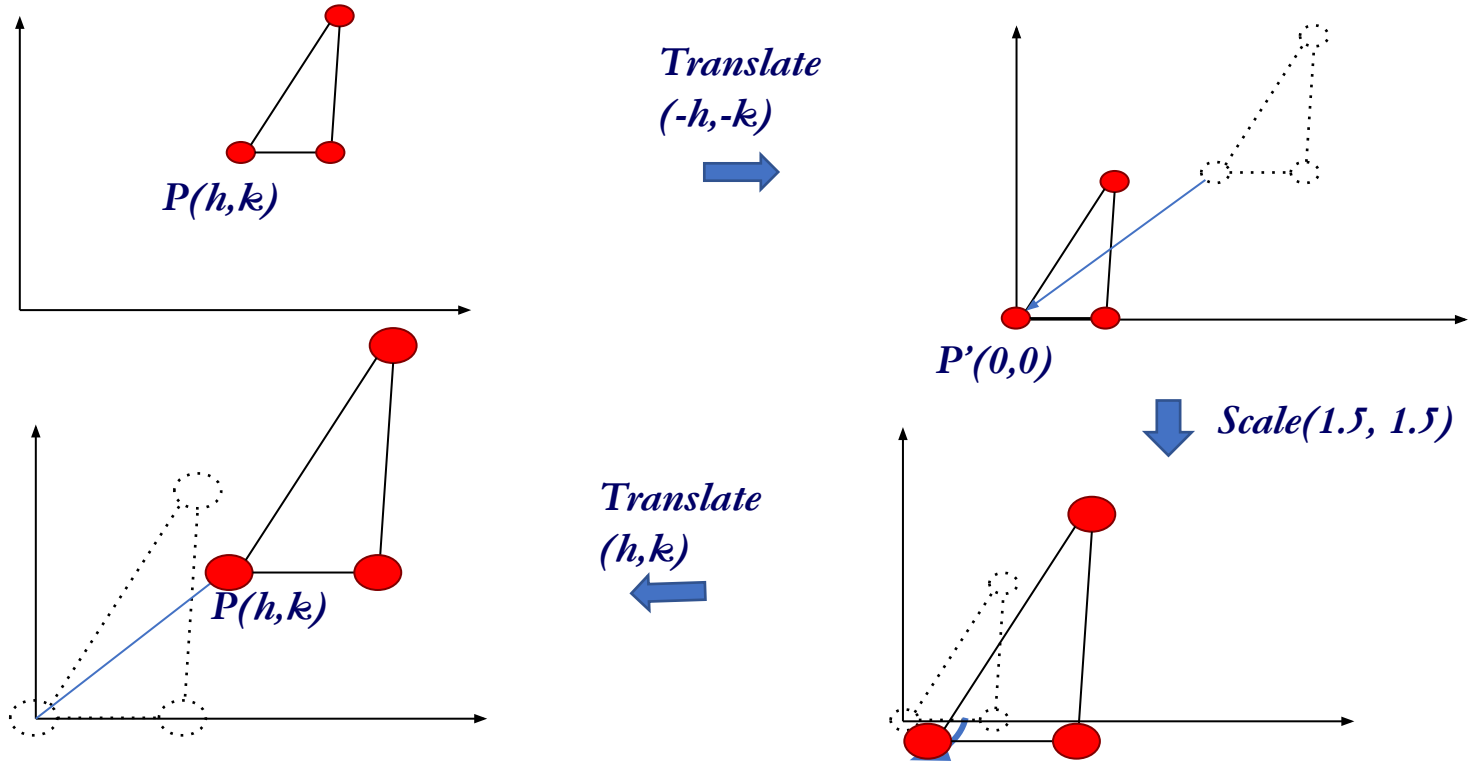
Scaling of $S_{a,b}$ about P(h,k): $R_{S,P}$

Step 1: Translate P(-h,-k) to origin

Step 2: Scale $S_{a,b}$ w.r.t to origin

Step 3: Translate (0,0) to P(h,k)

$$\mathbf{R_{S,P} = T(h,k) * S(a,b) * T(-h,-k)}$$


P(h,k)
Translate
(-h,-k)
P'(0,0)
Scale(1.5, 1.5)
Translate
(h,k)
P(h,k)

$$P' = T_{(h,k)} \times S_{(a,b)} \times T_{(-h,-k)} \times P$$

$$Composite\ matrix, M = T_{(h,k)} \times S_{(a,b)} \times T_{(-h,-k)}$$

$$= \begin{bmatrix} 1 & 0 & h \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -h \\ 0 & 1 & -k \\ 0 & 0 & 1 \end{bmatrix}$$

Example 2

Suppose, a composite transformation is defined as scaling 2 times in both axis with respect to point (5, 5). Calculate the composite transformation matrix in homogeneous form. Then, find the new coordinates of the point (10, 10) after transformation.

$$Composite\ matrix, M = T_{(5,5)} \times S_{(2,2)} \times T_{(-5,-5)}$$

$$= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & -5 \\ 0 & 2 & -5 \\ 0 & 0 & 1 \end{bmatrix}$$

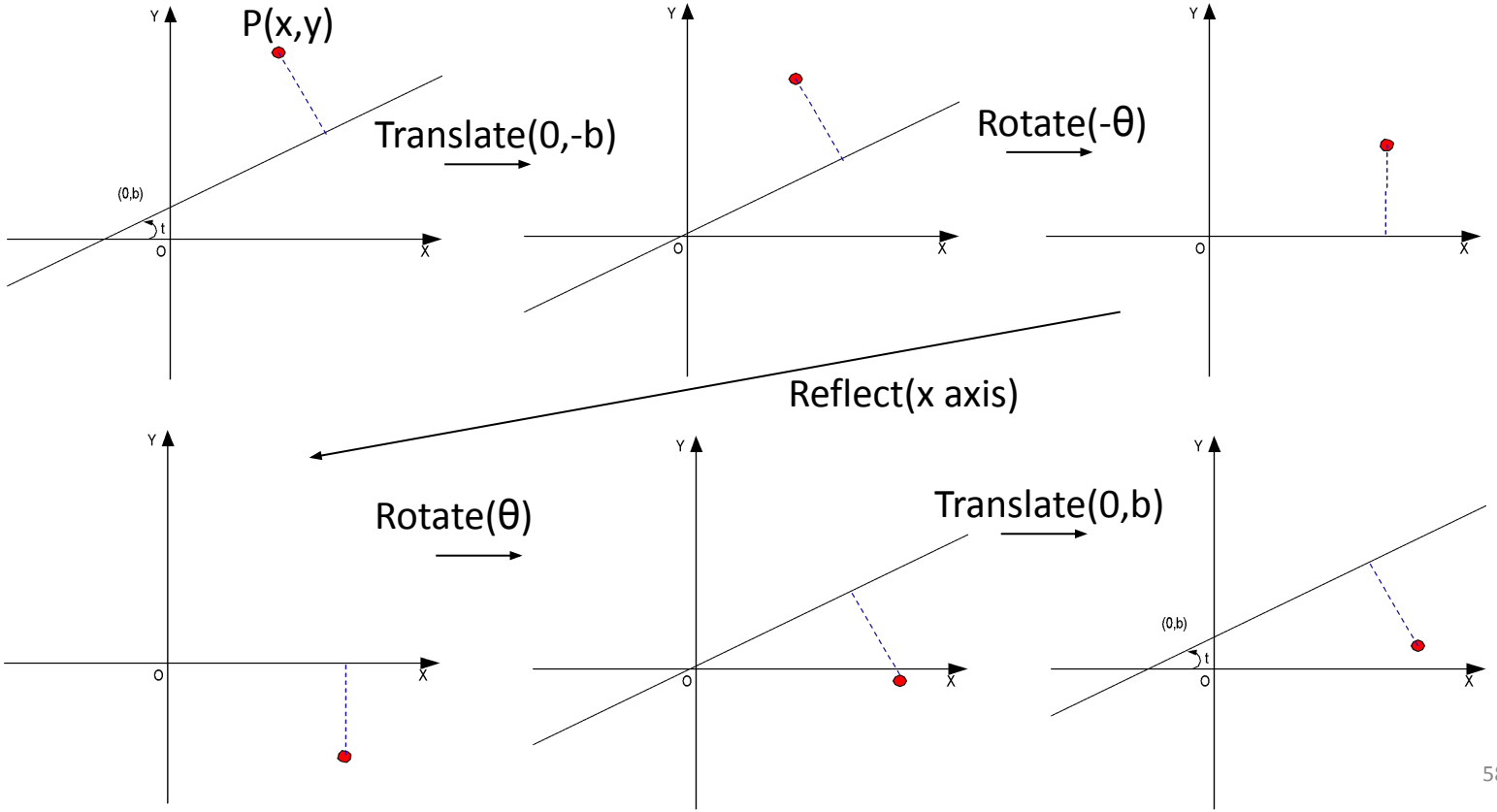
$$P' = M \times P$$

$$= \begin{bmatrix} 2 & 0 & -5 \\ 0 & 2 & -5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 15 \\ 15 \\ 1 \end{bmatrix}$$

So, the new coordinate of P is (15, 15)

Reflection about line L: y = mx +b



Reflection about line L: y = mx +b

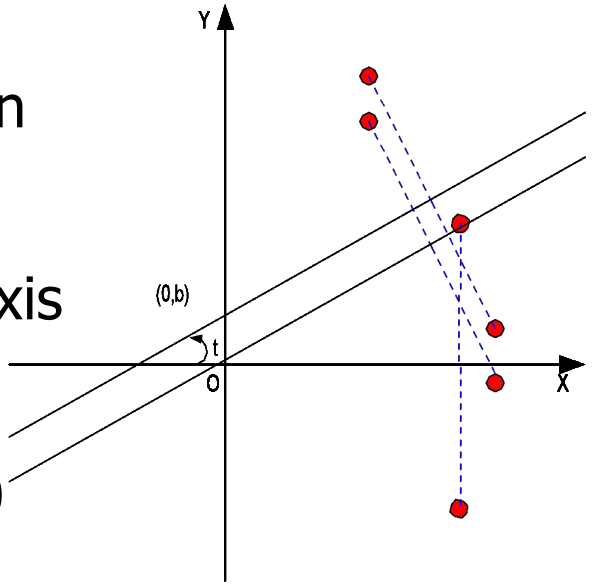
Step 1: Translate (0,-b) to origin

Step 2: Rotate -θ degrees

Step 3: Mirror reflect about X-axis

Step 4: Rotate θ degrees

Step 5: Translate origin to (0,b)



$$M_L = T(0, b) * R(\theta) * M_x * R(-\theta) * T(0, -b)$$

Reflect the point (10, 5) with respect to the line, y = x + 2

$$b = 2, m = 1, \theta = \tan^{-1} 1 = 45$$

$$M_{composite} = T_{(0,2)} \times R_{(45)} \times Refl_{(x\ axis)} \times R_{(-45)} \times T_{(0,-2)}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 45 & -\sin 45 & 0 \\ \sin 45 & \cos 45 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(-45) & -\sin(-45) & 0 \\ \sin(-45) & \cos(-45) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P' = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 12 \\ 1 \end{bmatrix}$$

New coordinate (3, 12)

Inverse Composite Transformations

1. A 3D point M is transformed to **M' (125, 255, -675)** by applying a **uniform scaling** with a factor of 5 with respect to a point (-33, 44, -55) followed by a counterclockwise **rotation** of 90^0 with respect to the same point (-33, 44, -55). What was the **old coordinate**, **M**?
2. Check Exercise Sheet: **Question 5(b)**

References

- Chapter 4 and Chapter 6, Schaum's Outline of Computer Graphics (2nd Edition) by Zhigang Xiang, Roy A. Plastock
- Chapter 5, Computer Graphics: Principles and Practice in C (2nd Edition) by James D. Foley, Andries van Dam, Steven K. Feiner, John F. Hughes
- Chapter 6, Fundamentals of Computer Graphics, by Peter Shirley